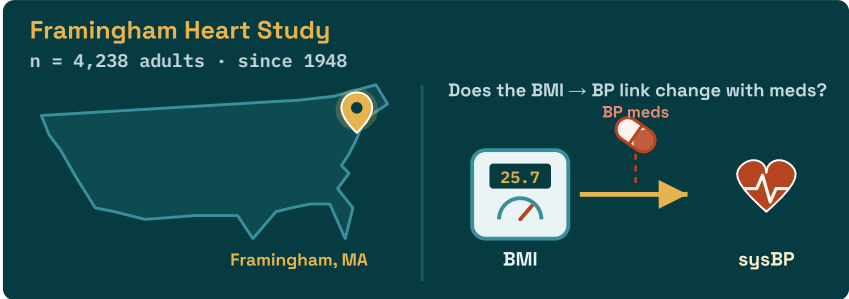


Interaction terms in regression

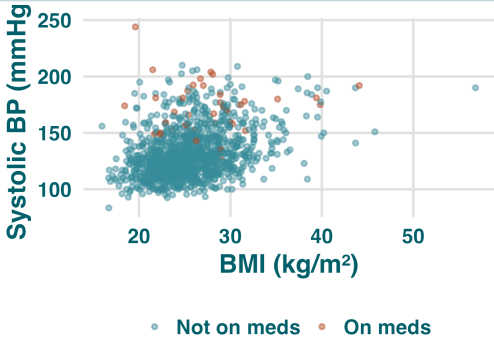
The study



```
# Framingham Heart Study, from LSTbook
dat <- LSTbook::Framingham |>
  select(sysBP, BMI, BPMeds, age, totChol,
         heartRate, cigsPerDay) |>
  mutate(BPMeds = factor(BPMeds,
    labels = c("Not on meds", "On meds")))

model_dat <- dat |>
  filter(!is.na(sysBP), !is.na(BMI), !is.na(BPMeds))

# colour the points by med status
ggplot(model_dat, aes(BMI, sysBP, colour = BPMeds)) +
  geom_point(alpha = .45)
```



People on BP meds have higher systolic BP, but does the BMI slope look the same in both groups?

What is an interaction?

An **interaction** means the effect of one predictor on the outcome **depends** on the level of another predictor. Graphically: the lines are **not parallel**.

NO INTERACTION
Same BMI slope in both groups; lines differ only by height.

INTERACTION
BMI slope differs by BP-med status; lines cross or diverge.

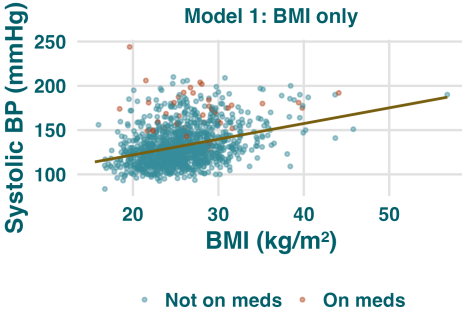
$$\text{sysBP} = \beta_0 + \beta_1 \text{BMI} + \beta_2 \text{meds} + \beta_3 (\text{BMI} \times \text{meds})$$

Model 1 · BMI only

```
m1 <- lm(sysBP ~ BMI, data = model_dat)
coef(summary(m1))[, c(1, 4)]
#>               Estimate Pr(>|t|)
#> (Intercept)  86.551    <2e-16
#> BMI          1.772    <2e-16
```

$$\widehat{\text{sysBP}} = 86.55 + 1.77(\text{BMI})$$

```
model_dat$fit <- predict(m1)
ggplot(model_dat, aes(BMI, sysBP)) +
  geom_point(alpha=.45) + geom_line(aes(y=fit))
```



Practice: Predict sysBP for someone on meds with BMI = 30

$$86.55 + 1.77(30) = \underline{\hspace{1cm}} \text{ mmHg} \quad \checkmark 139.7 \text{ mmHg}$$

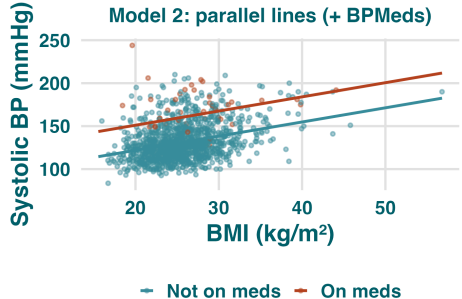
Meds are not in this model, so being on meds does not change the prediction.

Model 2 · add BP meds (parallel lines)

```
m2 <- lm(sysBP ~ BMI + BPMeds, data = model_dat)
coef(summary(m2))[, c(1, 4)]
#>               Estimate Pr(>|t|)
#> (Intercept)  88.842    <2e-16
#> BMI          1.650    <2e-16
#> BPMedsOn meds 29.240    <2e-16
```

$$\widehat{\text{sysBP}} = 88.84 + 1.65(\text{BMI}) + 29.24(\text{meds})$$

```
model_dat$fit <- predict(m2)
ggplot(model_dat, aes(BMI, sysBP, colour=BPMeds)) +
  geom_point(alpha=.45) + geom_line(aes(y=fit)) +
  scale_x_continuous(limits = c(0, NA))
```



Practice: Calculate the SysPB of a person A) On meds and B) Not on meds, with BMI 25

A) On meds: $88.84 + 1.65(25) + 29.24 = \underline{\hspace{1cm}}$ mmHg

B) Not on meds: $88.84 + 1.65(25) = \underline{\hspace{1cm}}$ mmHg ✓ A) 159.3 · B) 130.1

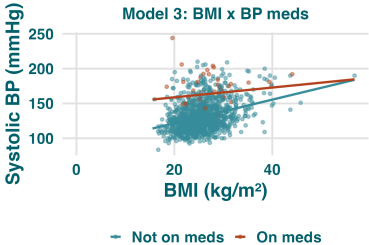
Model 3 · BMI x BP meds

```
m3 <- lm(sysBP ~ BMI + BPMeds + BMI:BPMeds, data = model_dat)
# Same as: lm(sysBP ~ BMI * BPMeds, data = model_dat)
coef(summary(m3))[, c(1, 4)]
#>               Estimate Pr(>|t|)
#> (Intercept)  87.644    <2e-16
#> BMI          1.696    <2e-16
#> BPMedsOn meds 57.106    3.7e-08
#> BMI:BPMedsOn meds -0.994    0.0062
```

$$\widehat{\text{sysBP}} = 87.64 + 1.70(\text{BMI}) + 57.11(\text{meds}) - 0.99(\text{BMI} \times \text{meds})$$

Reference	not on meds
87.64	intercept for not-on-meds group at BMI = 0
1.70	BMI slope for not-on-meds group
57.11	on-meds intercept shift at BMI = 0
-0.99	on-meds BMI slope adjustment: $1.70 - 0.99 = 0.70$

```
model_dat$fit <- predict(m3)
ggplot(model_dat, aes(BMI, sysBP, colour=BPMeds)) +
  geom_point(alpha=.45) + geom_line(aes(y=fit))
```



SLOPES BY GROUP

Not on meds:
 $\beta_1 = 1.70$
mmHg/BMI
On meds:
 $\beta_1 + \beta_3 = 0.70$
mmHg/BMI

Worked example · Predict sysBP for someone on meds, BMI = 30

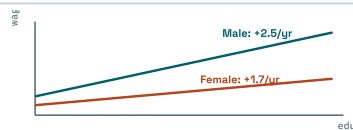
$$\widehat{\text{sysBP}} = 87.64 + 1.696(30) + 57.106(1) - 0.994(30 \times 1) \\ = 87.64 + 50.88 + 57.11 - 29.82 = 165.8 \text{ mmHg}$$

Practice: Predict sysBP for someone not on meds, BMI = 35

$$87.64 + 1.696(35) = \underline{\hspace{1cm}} \text{ mmHg} \quad \checkmark 147.0 \text{ mmHg}$$

Another example · wage ~ educ × sex

```
m <- lm(wage ~ educ + sex + educ:sex, data = dat)
coef(summary(m))[, c(1, 4)]
#>               Estimate Pr(>|t|)
#> (Intercept)    9.00    <.001
#> educ          1.70    <.001
#> sexMale       3.00    0.010
#> educ:sexMale  0.80    0.020
```



$$\text{wage} = 9.00 + 1.70(\text{educ}) + 3.00(\text{male}) + 0.80(\text{educ} \times \text{male})$$

Practice · male, 16 years of education

$$9.0 + 1.70(16) + 3.0 + 0.80(16) = \underline{\hspace{1cm}} \quad \checkmark 52.0$$

When to fit an interaction

Model 4 · all predictors, additive

```
m4 <- lm(sysBP ~ BMI + BPMeds + age +
        totChol + heartRate + cigsPerDay,
        data = model_dat)
```

#>	Estimate	Pr(> t)
#> (Intercept)	26.923	<2e-16
#> BMI	1.333	<2e-16
#> BPMedsOn meds	24.086	<2e-16
#> age	0.814	<2e-16
#> totChol	0.034	4.7e-07
#> heartRate	0.289	<2e-16
#> cigsPerDay	-0.011	0.665

$$\widehat{\text{sysBP}} = 26.92 + 1.33(\text{BMI}) + 24.09(\text{meds}) + 0.81(\text{age}) + 0.03(\text{chol}) + 0.29(\text{heartRate}) - 0.01(\text{cigs})$$

INTERPRET THE BMI EFFECT

The BMI coefficient is **1.333**. Holding medication status, age, cholesterol, heart rate, and smoking constant, each additional BMI unit is associated with about **1.33 mmHg higher** systolic BP. This one slope applies to everyone.

Model 5 · every pairwise interaction

```
m5 <- lm(sysBP ~ BMI + BPMeds + age + totChol + heartRate + cigsPerDay +
        BMI:BPMeds + BMI:age + BMI:totChol + BMI:heartRate + BMI:cigsPerDay +
        BPMeds:age + BPMeds:totChol + BPMeds:heartRate + BPMeds:cigsPerDay +
        age:totChol + age:heartRate + age:cigsPerDay +
        totChol:heartRate + totChol:cigsPerDay + heartRate:cigsPerDay,
        data = model_dat)
```

```
length(coef(m5))
#> [1] 22
```

```
coef(summary(m5))[, c(1, 4)]
```

#>	Estimate	Pr(> t)
#> (Intercept)	81.650	0.0002
#> BMI	0.570	0.3982
#> BPMedsOn meds	32.743	0.0896
#> age	0.099	0.7740
#> totChol	-0.042	0.5561
#> heartRate	-0.228	0.2801
#> cigsPerDay	0.251	0.3587
#> BMI:BPMedsOn meds	-1.054	0.0028
#> BMI:age	0.008	0.3526
#> BMI:totChol	0.000	0.8101
#> BMI:heartRate	0.004	0.4790
#> BMI:cigsPerDay	0.001	0.8288
#> BPMedsOn meds:age	0.224	0.3343
#> BPMedsOn meds:totChol	0.044	0.2499
#> BPMedsOn meds:heartRate	-0.056	0.6440
#> BPMedsOn meds:cigsPerDay	0.220	0.2333
#> age:totChol	0.000	0.6195
#> age:heartRate	0.006	0.0529
#> age:cigsPerDay	-0.004	0.2445
#> totChol:heartRate	0.001	0.2693
#> totChol:cigsPerDay	-0.000	0.7089
#> heartRate:cigsPerDay	-0.001	0.6731

INTERPRET THE BMI EFFECT

There is no single BMI coefficient any more. The BMI slope is built from every row BMI appears in, and each interaction is **multiplied by that person’s own value**:

$$\text{slope} = 0.570 - 1.054(\text{meds}) + 0.008(\text{age}) + 0.000(\text{chol}) + 0.004(\text{HR}) + 0.001(\text{cigs})$$

WORKED SLOPE: UNMEDICATED, AGE 60, HR 75

$$0.570 + 0.008(60) + 0.004(75) \approx 1.35$$

A different age or heart rate gives a different BMI slope, so every person effectively has their own. No single number answers “what does one more BMI unit do?”

When should you fit an interaction?

- 1 · ESTABLISHED IN THE LITERATURE**

Other studies have reported this interaction as a stable finding, so you include it to match the standard model in your field. Example: age times sex in growth curves.
- 2 · IT IS THE RESEARCH QUESTION**

You are running the study to ask whether the effect of X on Y depends on Z. The interaction coefficient is the estimate you care about, so you fit it and test it.
- 3 · THERE IS A DESIGN OR MECHANISM REASON**

The study design or biology strongly suggests that one effect should differ across groups. Example: an antibiotic lowers infection severity in non-resistant strains, but has little effect on resistant strains.
- 4 · OTHERWISE, ADDITIVE IS OFTEN A REASONABLE DEFAULT**

If there is no specific reason to expect effect modification, an additive model is often easier to interpret and more stable..